

Bluetooth Control Lab · Premium Course

2 live sessions · 2 hrs each · 11-15 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Control an Arduino gadget wirelessly from a phone over Bluetooth, then build a custom app to command it.

DETAIL

Track	Arduino Robotics & Electronics
Format	2 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	11-15 yrs
Level	Intermediate → Advanced
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16x2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 1,000: HC-05 Bluetooth module, Small DC motor
Standalone	No
Prerequisites	Button controlled LED (free, recommended)

Programme

1. **Connect & command.** Wire an HC-05 Bluetooth module, pair it with a phone, and send commands from a terminal app to switch LEDs and drive a motor — understanding the serial link underneath.

2. **Build your app.** Use MIT App Inventor to make a custom phone app with your own buttons, and send live sensor readings (distance or temperature) back to the phone for true two-way communication.

What they take home

 Bluetooth-controlled gadget + their own phone app

Why parents love it

“Their phone becomes the remote control — they wire the hardware, build the app, and make the two talk to each other.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Bluetooth Control Lab**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.